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Increasing doses of biocholine on apparent digestibility, ruminal fermentation, and performance in dairy cows

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HIGHLIGHTS

- Increasing doses of biocholine (BC) were tested in mid-lactation cows
- BC had no effect on nutrient intake, but linearly increased fat digestibility
- BC did not affect ruminal total volatile fatty acid concentration
- BC had a quadratic positive effect on milk yield

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ABSTRACT

Biocholine is an herbal feed additive considered a natural alternative to synthetic choline, with relatively low rumen degradability composed of cellulose and choline conjugates, namely phosphatidylcholine. Rumen-protected choline has been largely evaluated during the peripartum period, but few studies evaluated the effects of choline supplementation after the transition period. The objective of this study was to determine the effects of increasing doses of biocholine on feed intake, feed particle sorting, total apparent digestibility, ruminal fermentation, milk yield and composition, purine derivative excretion, and nitrogen balance. Twenty-four Holstein cows (163 ± 98 days in milk, 27.6 ± 7.14 kg/d milk yield), 4 of which were ruminally cannulated, were enrolled to this study. Cows were blocked according to milk yield, days in milk, and presence of ruminal cannulas, then all cows were randomly assigned to a treatment sequence in a replicated 4×4 Latin square experiment design. The experimental treatments were a control (CON) diet, and biocholine (BC; Biocholine FC®, Nutriquest Technofeed Nutrição Animal Ltda., Campinas, Brazil) fed at 7, 14, and 21 g/d. Cows were allowed 14 d of adaptation to treatments and the last 7 days of each period were used for sampling. The forage to concentrate ratio of the basal diet was 48:52 (on a dry matter basis) with corn silage as the single forage source, and diet was provided in the form of total mixed ration. No treatment effects were detected on feed intake, but BC quadratically affected ($P = 0.010$) sorting for feed particles >19 mm and linearly increased ($P = 0.037$) ether extract digestibility (840, 834, 843, and 853 g/kg for cows fed BC at 0, 7, 14, and 21 g/d, respectively). Biocholine supplementation tended to quadratically decrease ruminal pH (6.17, 6.04, 6.11, and 6.11 for cows fed BC at 0, 7, 14, and 21 g/d, respectively) but had no effect on concentrations of total volatile fatty acids and $\text{NH}_3\text{-N}$. Biocholine had a positive quadratic effect ($P \leq 0.042$) on milk yield (29.5, 30.5, 30.0, and 29.8 kg/d for cows fed BC at 0, 7, 14, and 21 g/d, respectively) and feed efficiency (1.20, 1.25, 1.21, and 1.21 kg milk/kg DM intake for cows fed BC at 0, 7, 14, and 21 g/d, respectively). Biocholine tended to quadratically affect ($P = 0.058$) N excretion through milk (157, 162, 160, and 159 g/d for cows fed BC at 0, 7, 14, and 21 g/d, respectively). There was no evidence for treatment effects on purine derivatives excretion. In conclusion, feeding BC may improve

Abbreviations: BC, biocholine; BW, body weight; CON, control; CP, crude protein; DIM, days in milk; DM, dry matter; DMI, dry matter intake; ECM, energy corrected milk; EE, ether extract; FCM, fat-corrected milk; iNDF, indigestible NDF; aNDF, neutral detergent fiber treated with alpha amylase inclusive ash; PD, purine derivatives; TMR, total mixed ration; VFA, volatile fatty acids.

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milk yield and feed efficiency of dairy cows. Predicted maxima response in milk yield and feed efficiency were observed at 10 g/d of BC.

Introduction

Choline, or trimethyl ethanolamine, is a strong base with a trimethylated quaternary nitrogen. Choline is critical for synthesis of acetylcholine and phosphatidyl molecules. The former is an important neurotransmitter in parasympathetic nervous system (Pinotti et al., 2002) and the latter is essential to maintaining the structure of cellular membrane (Davidson et al., 2008), synthesis and secretion of very-low-density lipoprotein in the liver (Cooke et al., 2007), and synthesis and secretion of milk (Kinsella, 1969). Choline deficiency in mammals is rarely identified because other molecules may contribute with choline requirements, including methionine, betaine, folate, and vitamin B12 (Zeisel, 1988).

Dietary choline has a minor contribution to the pool of choline in adult ruminants as choline in feedstuffs is extensively degraded in the rumen (79.4% to 83.8%) and synthetic choline chloride is even more degraded (98.6%) in the rumen (Jayaprakash et al., 2016). In addition, choline chloride reduces the stability of vitamins in premixes (Coelho, 2002). In dairy cows, especially at the onset of lactation, the excretion of methylated products is relatively high while the intake of methionine is generally limited when compared to dry or late-lactation cows (Baldi and Pinotti, 2006). The total choline concentration in plasma increases almost 10-times (1,305 $\mu\text{mol/L}$ at the 1st wk and 11,536 $\mu\text{mol/L}$ at the 37th wk of lactation) from early to late lactation; the yield of phosphatidylcholine also increases over the lactation (Artegoitia et al., 2014). These authors also found that phosphatidylcholine (as % of total choline forms) was the main choline metabolite in blood after early lactation and increased from 36% at week 7 to 71% at week 37 of lactation (Artegoitia et al., 2014). Phosphatidylcholine can be synthesized in the body through phosphorylation of dietary choline or endogenously mediated by phosphatidylethanolamine N-methyltransferase in the liver (Reo et al., 2002) and mammary gland (Chao et al., 1988).

Because of its ability to limit lipolysis (Cooke et al., 2007), improving export of hepatic and uptake of fatty acids (Piepenbrink and Overton, 2003; Zenobi et al., 2018), or decreasing inflammation (Sun et al., 2016), most research with choline-compounds were dedicated to peripartum dairy cows, whereas few studies have been performed with mid- or late-lactation cows. Increasing dietary choline availability during the transition period and early lactation has improved milk production (Baldi and Pinotti, 2006; Brusemeister and Sudekum, 2006), and the metabolism of lipids (Cooke et al., 2007; Piepenbrink and Overton, 2003) and methyl group (Baldi and Pinotti, 2006). Furthermore, studies have shown evidence that rumen-protected choline supplementation to transition cows can reduce inflammation (Zenobi et al., 2020) and improve antioxidant capacity (Sun et al., 2016). Regarding mid-lactation or cows after the peak of lactation, either the abomasal infusion of choline or feeding increasing doses of rumen-protected choline have improved milk yield without affecting dry matter intake (DMI; Erdman and Sharma, 1991; Sharma and Erdman, 1989). Recently, Holdorf and White (2021) in a heat stress study using mid-lactation cows [208 $\pm$ 31 days in milk (DIM)] found that feeding rumen-protected choline tended to increase milk yield during the baseline period (3 days) and tended to ameliorate the loss of milk yield under heat stress. Because of the tendency for improved milk yield after 3 days of supplementation, these authors suggested that rumen-protected choline supplementation to mid-lactation cows should be further explored. Altogether, these experiments conflict with results reported by Bollatti et al. (2020) that no improvements in milk yield were observed when feeding rumen-protected choline after the transition period.

The rumen-protected choline most fed to dairy cows is currently marketed as choline chloride, a synthetic choline protected by

microencapsulation with 87% of bioavailability as described by Benoit (2009). Biocholine (BC; Biocholine FC®, Nutriquest Technofeed Nutrição Animal Ltda., Campinas, Brazil) is an herbal product containing phosphatidylcholine extracted from plants (*Azadirachta indica*, *Citrillus colocynthis*, *Trachyspermum ammi*, and *Achyranthes aspera*) which has emerged as an alternative to the choline sources marketed to livestock. Feeding BC to feedlot lambs linearly increased the feed conversion rate and tended to increase blood phosphatidylcholine concentration (Martínez-Aispuro et al., 2019). Supplementing ewes with BC at 5 g/d during the pre-partum period and early lactation tended to improve milk yield compared to the control group (Alba et al., 2020). To the best of our knowledge, there is no randomized controlled trial that evaluated BC supplementation to dairy cows. In the present study we sought to investigate the influence of increasing dietary doses of BC on feed intake, nutrient digestibility, ruminal fermentation, milk yield and composition, and nitrogen utilization in mid-lactation cows. We hypothesized that feeding BC to mid-lactation cows would improve milk yield in a dose dependent manner, with minor effects on ruminal fermentation.

Materials and methods

This experiment was carried out under approval of the Ethics Committee on Animal Use – School of Veterinary Medicine and Animal Sciences, University of Sao Paulo (Sao Paulo, Brazil; protocol #5899210319).

Cows, experimental design, and treatments

Twenty-four Holstein cows [163 $\pm$ 98 DIM, 27.6 $\pm$ 7.14 kg/d milk yield, 636 $\pm$ 83.2 kg body weight (BW), 2.59 $\pm$ 0.154 body condition score, and 2.08 $\pm$ 0.78 parturition at the beginning of experiment], 4 of which were ruminally cannulated, were enrolled in this study. Cows were housed in a barn with individual pens (17.5 m² area), sand bedding, fans, feed bunks, and free access to water. Cows were blocked according to milk yield, DIM, and presence of rumen cannulas, then all cows were randomly assigned to a treatment sequence in a 4 $\times$ 4 Latin square design. The experimental treatments were a control (CON) diet, and biocholine (BC; Biocholine FC®, Nutriquest Technofeed Nutrição Animal Ltda.) targeting a feeding rate of 7, 14, and 21 g/d. Biocholine was mixed with ground corn (1:8 w/w on wet basis), then this mixture was incorporated with micro and macrominerals of concentrate. Afterwards, the latter mixture was mixed with the rest of concentrate ingredients at the feed mill. Concentrate was prepared weekly during the experiment. Cows were allowed 14 d of adaptation to treatments and the last 7 days of each period were used for data collection and sampling. According to manufacturer's information, the feed additive is a green powder with herbal odor containing a minimum of 16 g/kg phosphatidylcholine. Wet chemistry analysis revealed the following chemical composition of BC: 933 g/kg DM, 431 g/kg neutral detergent fiber (aNDF), 317 g/kg acid detergent fiber, 69.0 g/kg lignin (sa), 94.6 g/kg crude protein (CP), and 54.7 g/kg ether extract (EE). Cows were fed a total mixed ration (TMR) twice daily (0700 and 1300 h) targeting refusals between 5 and 10% (as-fed basis). The basal diet had a forage to concentrate ratio of 48:52 and corn silage was used as the single forage source. Concentrate ingredients were ground corn (183 g/kg DM), soybean meal (138 g/kg DM), citrus pulp (82 g/kg DM), whole raw soybean (64.7 g/kg DM), bypass soybean meal (21.2 g/kg DM) and minerals (31 g/kg DM). Diet (Table 1) was formulated according to NRC (2001) using the following inputs: 600 kg BW, 30 kg/d milk yield, 3.5% milk fat, 3.2% milk protein, and 4.85% milk lactose.

Table 1

Ingredients and chemical composition of experimental diet (g/kg DM, unless otherwise stated).

Item	Diet
Ingredient	
Corn silage	480
Ground corn	183
Soybean meal, 48% CP	138
Citrus pulp	82.0
Whole raw soybean	64.7
Bypass soybean meal ¹	21.2
Sodium bicarbonate	8.20
Limestone	7.30
Salt	2.60
Minerals ²	12.9
Chemical composition	
Dry matter (as-fed)	460
Non-fiber carbohydrate ³	426
Starch	228
Crude Protein	167
Rumen degradable protein ⁴	104
Rumen undegradable protein ⁴	60.0
Neutral detergent fiber	322
Acid detergent fiber	173
Ether Extract	30.4
Total digestible nutrient ⁴	686
NE _L (3 ×), Mcal/kg ⁴	1.56
Particle size distribution, % as-fed	
>19 mm	10.8
19–8 mm	39.4
8–4 mm	17.8
<4 mm	32.0

1Composition (per kg): 215 g Ca, 60 g P, 20 g S, 20 g Mg, 35 g K, 70 g Na, 15 mg Co, 700 mg Cu, 600 mg F, 10 mg Cr, 700 mg Fe, 40 mg I, 1,600 mg Mn, 20 mg Se, 2,500 mg Zn, 200,000 IU Vit A, 50,000 IU Vit D3, and 1,500 IU Vit. E.

2Soypass (Cargill, Wayzata, MN).

3According to Hall (2000).

4Calculated according to NRC (2001).

Feed intake, apparent digestibility, and sorting index

Samples of corn silage and refusals were collected daily during the sampling period and pooled into a composite sample per period. Samples from concentrate ingredients were collected weekly at the feed mill during concentrate preparation and pooled into a composite sample per period. Corn silage moisture was assessed twice weekly for diet adjustments when necessary. Samples were partially dried in a forced-air oven at 55°C for 72 h and ground at 1 mm in a Wiley mill (Arthur H. Thomas, Philadelphia, PA). The contents of DM (method 930.15; AOAC International, 2000), ash (method 942.05; AOAC International, 2000), organic matter (DM – ash), CP (N × 6.25; Kjeldahl method 984.13; AOAC International, 2000), and EE method 920.39; AOAC International, 2000) in samples were evaluated. Neutral detergent fiber (Van Soest et al., 1991) was analyzed using α-amylase and sodium sulfite to the detergent (TE-149 fiber analyzer; Tecnal Equipamentos para Laboratório Inc., Piracicaba, Brazil), and ADF and lignin (method 973.18) were analyzed according to AOAC (2000). Non-fiber carbohydrate was calculated according to Hall (2000), and total-digestible nutrient was calculated according to NRC (2001).

Daily fecal DM excretion of cows was estimated using indigestible neutral detergent fiber (iNDF) as an internal marker. Fecal samples (n = 9, ~100 g each) were collected every 9 h during 3 consecutive days of each experimental period (d 17, 18, and 19), and pooled into a composite sample per animal. Feed, refusals, and fecal partially dried samples were ground to pass through a 2-mm screen in a Wiley mill (Arthur H. Thomas) and were placed in nonwoven fabric bags (5 × 5 cm and 20 mg of DM/cm² surface; Casali et al., 2008). Samples (quadruplicate) were incubated in the rumen of two cannulated dry cows for 288 h. After incubation, samples were washed in running tap water until the rinse

water became clear and then analyzed for NDF (Van Soest et al., 1991) to obtain the iNDF content (Huhtanen et al., 1994).

The DM and nutrient intake (NI; kg/d) were calculated by the difference between nutrient offered (kg/d) and refused (kg/d). The total apparent digestibility of DM and nutrients was calculated as follows:

$$DM \text{ digestibility (g / kg)} = \frac{[DMI - FE] \times 1000}{DMI},$$

$$Nutrient \text{ digestibility (g / kg)} = \left[\frac{NI - NE}{NI} \right] \times 100,$$

where, FE is the fecal excretion calculated as follows:

$$FE \text{ (kg / d)} = \frac{iNDF \text{ intake (kg/d)}}{iNDF \text{ in feces (kg/kg)}},$$

and nutrient excretion (NE; g/kg) is $FE \text{ (kg/d)} \times NCF \text{ (g/kg)}$, where NCF is nutrient concentration in feces.

Samples of TMR and refusals were collected during 2 consecutive days and assessed for particle size distribution (Penn State Particle Separator, Nasco, Fort Atkinson, WI, USA). Sorting index was calculated according to Silveira et al. (2007), where values greater than 1 correspond to sorting for a specific particle size, values lower than 1 means sorting against a specific particle size, and values equal to 1 means no sorting. The sorting index (SI) was calculated as the actual intake divided by the expected intake for each portion retained on the individual sieves according to Silveira et al. (2007):

$$SI = \frac{(FO \times PS_{TMR}) - (FR \times PS_O)}{FI \times PS_{TMR}},$$

where: FO is the amount of feed offered, PS_{TMR} is the particle size distribution of TMR (% as fed basis), FR is the feed amount refused, PS_O is the particle size distribution from the refused samples (% as fed basis), and FI is the actual intake (as-fed basis).

Ruminal fermentation

Rumen fluid samples were collected from cannulated cows on the last day of each period, before the morning feeding (0h), and after 3, 6, 9, 12 and 15 h. Digesta were collected from different sites within the rumen (dorso-cranial, ventral-cranial, ventral, caudo-ventral and caudo-dorsal regions) and strained through 4 layers of cheesecloth to extract rumen fluid (250 mL). Rumen fluid pH was measured by a glass electrode and a reference electrode (MB-10, Marte Científica®, Santa Rita do Sapucaí, Brazil). Samples of rumen fluid were acidified with formic acid (5:1 v/v) and stored frozen for further volatile fatty acids (VFA) and NH₃-N analyses. Rumen fluid samples were thawed, centrifuged (500 × g for 15 min, at room temperature) and an aliquot from the supernatant (2 mL) was collected and mixed with a sulfuric acid solution (1 mL at 1 N) for NH₃-N analysis using the phenol-hypochlorite method (Broderick and Kang, 1980). Absorbances were measured on a microplate reader (Biochrom Asys Microplate Reader, Biochrom®, Cambridge, UK). Another aliquot from the supernatant was mixed with formic acid (at a ratio of 4 to 1) for VFA analysis according to Ferreira et al. (2018). Gas chromatograph, capillary column, and chromatographic run settings are described in (Zilio et al., 2019).

Milk composition and serum glucose

Cows were milked twice a day (0600 and 1600 h) and milk weights were electronically recorded (Alpro®, DeLaval, Tumba, Sweden). Milk samples (300 mL) were collected during 3 consecutive days (d 15, 16, and 17 of each experimental period) from all milkings. Milk samples collected 0600 h were stored in the fridge until pm milk samples were collected. Milk samples from both shifts were pooled by cow proportionally to the yield and analyzed for fat, protein, and lactose by mid

infrared method (Lactoscan®, Entelbra, Londrina, Brazil). Fat-corrected milk (3.5%; FCM) and energy-corrected milk (ECM) were calculated according to Sklan et al. (1992) and Tyrrell and Reid (1965), respectively. An aliquot (10 mL) of milk samples was deproteinized according to Broderick and Clayton (1997) and analyzed for urea concentration using commercial kits (catalog no. K-082; Bioclin, Belo Horizonte, Brazil).

Blood samples were collected into vacutainers without clot activator (BD Vacutainer®, Becton, Dickinson and Company, Franklin Lakes, NJ) by puncture of coccygeal vessels on d 16 of each period, 4 h after the morning feeding. Blood tubes were centrifuged ($2,000 \times g$ for 15 min at room temperature), then serum was assessed for concentrations of glucose using a colorimetric kit (catalog no. K-082, Bioclin) and absorbance measured on a spectrophotometer (SBA 200, Celm, Barueri, Brazil).

Nitrogen excretion and microbial protein synthesis

Nitrogen intake was calculated as CP intake divided by 6.25 and nitrogen in milk was calculated as milk protein yield divided by 6.38. Nitrogen in feces and urine samples were determined (Kjeldahl method 984.13) according to AOAC International (2000). The excretion of purine derivatives (PD) in urine (uric acid and allantoin) and milk (allantoin) were determined according to Chen and Gomes (1992) and considered as a proxy of ruminal microbial protein supply (Shingfield, 2000). Urine (20 mL) samples were collected at the same time points of feces and were pooled into a composite sample per cow. Urine samples were diluted (1:4 ratio) into a sulfuric acid solution at 0.036 N (Chen and Gomes, 1992) to preserve PD. Samples were stored frozen for total nitrogen, allantoin, uric acid, and creatinine analyses. Daily urine volume was estimated considering a daily creatinine excretion of 24.05 mg/kg BW (Chizzotti et al., 2008). Urine creatinine concentration was assessed using commercial kits (kinetic creatinine catalog no. K-067, Bioclin) and absorbances measured on spectrophotometer (SBA 200, Celm). Allantoin concentrations in urine and milk were assessed according to Fujihara et al. (1987). Uric acid concentration in urine was analyzed using a commercial kit (uric acid stable liquid; catalog no. K-052; Bioclin). Total PD was calculated as the sum of allantoin and uric acid excreted in milk and urine (Boero et al., 2001). Body weight was measured on d 6, 7, 20, and 21 of each experimental period, before the morning feeding and after milking, using a digital scale for large animals (AWS100, DeLaval, Tumba, Switzerland).

Statistical analysis

Data were submitted to ANOVA using the PROC MIXED of SAS 9.4 (Statistical Analyses System for Windows - Institute Inc., Cary, USA) according to the following model:

$$Y_{ijkl} = \mu + A_i + P_j + S_k + AS_{ik} + \alpha_{kl} + \varepsilon_{ijkl}$$

where $\alpha_{kl} \sim N(0; \sigma_c^2)$ and $\varepsilon_{ijkl} \sim N(0; \sigma_e^2)$; where: Y_{ijkl} = observation on animal l , given treatment i , at period j , in square k ; μ = overall mean, A_i = fixed effect of the i th treatment ($i = 1$ to 4); P_j = fixed effect of the j th period ($j = 1$ to 4); S_k = fixed effect of the k th Latin square ($k = 1$ to 6); AS_{ik} = interaction fixed effect between treatment and Latin square; α_{kl} = random effect of animal within square ($l = 1$ to 24); ε_{ijkl} = random error associated with each observation; N = Gaussian distribution; σ_c^2 = estimated variance associated to cows; and σ_e^2 = estimated residual variance.

Ruminal fermentation variables were analyzed as repeated measures using the PROC MIXED of SAS, according to the following model:

$$Y_{ijklm} = \mu + A_i + P_j + B_k + \alpha_{kl} + \omega_{ijkl} + T_m + T \times A_{mi} + \varepsilon_{ijklm}$$

where $\omega_{ijkl} \sim N(0; \sigma_\omega^2)$, and $\varepsilon_{ijklm} \sim MVN(0; R)$; where Y_{ijklm} =

observation on animal l , given treatment i , at period j , in square k , in time m ; μ , A_i , P_j , B_k , and α_{kl} were previously defined; ω_{ijkl} = residual error associated with cows within experimental period; T_m = fixed effect of sampling time ($m = 1$ to 6); $T \times A_{mi}$ = fixed interaction effect between time and treatment; ε_{ijklm} = random residual error; MVN = multivariate normal; and R = variance-covariance matrix of residuals due to the repeated measurements. Auto regressive covariance matrices were evaluated [AR(1) and ARH(1)] and chosen according to the lowest BIC values.

Dose responses were evaluated through linear and quadratic contrasts and estimates of regression function for performance variables with quadratic effect were obtained using SOLUTION option and modeling square and period as random effects. A contrast between CON and BC supplementation was also assessed. Significance level was set at $P \leq 0.05$, and tendency was considered as $0.05 < P \leq 0.10$.

Results

Nutrient intake, sorting index, and digestibility

Treatments did not affect ($P > 0.481$) intake of DM, OM, CP, NDF (either kg/d or %BW), EE, and NFC of cows (Table 2). Increasing levels of BC had a positive quadratic effect ($P = 0.012$) on sorting for feed particles >19 mm, whereas BC treatments had a negative quadratic effect ($P = 0.003$) on sorting for feed particles between 19–8 mm size. Although no treatment differences were detected on digestibility coefficient of DM, CP, and NDF, feeding BC linearly increased ($P = 0.027$) EE digestibility and tended to quadratically affect ($P = 0.061$) OM digestibility.

Ruminal fermentation

Feeding BC tended to quadratically affect ($P = 0.085$) ruminal pH with values of 6.17, 6.04, 6.11, and 6.11 for cows fed BC fed at 0, 7, 14, and 21 g/d, respectively (Table 3). Treatments had no effect on ruminal concentrations of $\text{NH}_3\text{-N}$ and total VFA, as well as on the proportion of acetate, propionate, isobutyrate, butyrate, isovalerate, and valerate.

Milk yield and composition

Biocholine had a positive quadratic effect on milk yield ($P = 0.042$) and feed efficiency (milk yield $\div$ DMI; $P = 0.041$; Table 4). Furthermore, regression coefficient estimates revealed that the best dose of BC for optimization of milk yield and feed efficiency were 10.98 and 9.94 g/d, respectively. Feeding BC tended to show a positive quadratic effect on protein ($P = 0.089$) and lactose yields but tended to show a negative quadratic effect ($P = 0.095$) on milk fat content. Incrementing doses of BC tended to linearly increase ($P = 0.066$) serum glucose concentration. No treatment differences were detected in ECM, FCM, contents of protein and lactose, and milk urea nitrogen concentration.

Nitrogen and purine derivative excretion

Biocholine treatments did not affect nitrogen excretion (through urine and feces) but tended to show a positive quadratic effect ($P = 0.078$) on N excreted in milk (Table 5). There was no evidence for treatment effects on PD excretion including allantoin and uric acid in urine, and allantoin in milk.

Discussion

We hypothesized that feeding BC would improve the performance of mid-lactation cows with minor effects on ruminal fermentation. Indeed, this experiment demonstrated that BC may improve milk yield and feed efficiency. Although no effects were observed on nutrient intake, feeding

Table 2

Feed intake, total apparent digestibility, and sorting index of mid-lactation dairy cows fed increasing levels of biocholine (n = 24).

Item	Treatment ¹				SEM	P-value ²	
	CON	BC7	BC14	BC21		L	Q
Intake, kg/d							
Dry matter	24.9	24.7	25.0	25.0	0.35	0.634	0.623
Organic matter	23.5	23.4	23.8	23.6	0.33	0.619	0.924
Crude protein	4.22	4.20	4.20	4.22	0.055	0.981	0.667
NDF ³	7.73	7.71	7.83	7.79	0.110	0.481	0.900
NDF ³ , % BW	1.24	1.25	1.25	1.26	0.013	0.535	0.875
Ether extract	0.784	0.774	0.786	0.782	0.0107	0.863	0.758
Non-fiber carbohydrate	10.8	10.8	10.9	10.9	0.15	0.646	0.937
Body weight	640	644	642	642	7.5	0.833	0.265
Sorting index ⁴							
>19 mm	0.932	0.963	0.954	0.926	0.4771	0.577	0.012
19-8 mm	0.983	0.977	0.979	0.986	0.0013	0.303	0.003
8-4 mm	0.987	0.984	0.985	0.988	0.0014	0.676	0.265
<4 mm	1.047	1.047	1.048	1.050	0.0015	0.466	0.702
Digestibility coefficient, g/kg							
Dry matter	701	692	693	699	29.2	0.914	0.135
Organic matter	723	709	714	721	37.1	0.952	0.061
Crude protein	709	695	704	709	42.4	0.747	0.180
NDF ³	520	515	511	534	55.2	0.348	0.123
Ether extract	840	834	843	853	44.0	0.027	0.115

1Control (CON) and biocholine (Biocholine FC®, Nutriquest Technofeed Nutrição Animal Ltda., Campinas, Brazil) targeting a feeding rate of 7, 14, and 21 g/d (BC7, BC14, and BC21, respectively).

2P-values for contrasts: linear (L), quadratic (Q), and control vs biocholine treatments (CON vs BC).

3Neutral detergent fiber.

4Values greater than 1 mean that animals sorted for a particular particle size and values lower than 1 mean that animals sorted against a particular particle size (Silveira et al., 2007).

Table 3

Ruminal fermentation of mid-lactation dairy cows fed increasing levels of biocholine (n = 4).

Item	Treatment ¹				SEM	P-value ²	
	CON	BC7	BC14	BC21		L	Q
pH	6.17	6.04	6.11	6.11	0.033	0.423	0.085
NH ₃ -N, mg/dL	13.7	17.6	11.7	15.8	1.21	0.938	0.931
VFA, %							
Acetate	60.6	60.4	59.5	60.7	0.76	0.884	0.262
Propionate	19.3	19.1	20.3	19.5	0.45	0.217	0.314
Isobutyrate	1.25	1.18	1.15	1.18	0.037	0.444	0.460
Butyrate	14.3	14.8	14.5	14.1	0.44	0.652	0.422
Isovalerate	2.50	2.51	2.44	2.44	0.112	0.613	0.962
Valerate	2.01	2.02	2.11	2.10	0.081	0.349	0.884
Total VFA, mM	126	130	112	112	6.1	0.398	0.994
Acetate to propionate ratio	3.21	3.19	3.01	3.17	1.574	0.276	0.141

1Control (CON) and biocholine (Biocholine FC®, Nutriquest Technofeed Nutrição Animal Ltda., Campinas, Brazil) targeting a feeding rate of 7, 14, and 21 g/d (BC7, BC14, and BC21, respectively).

2P-values for contrasts: linear (L), quadratic (Q), and control vs biocholine treatments (CON vs BC).

BC quadratically increased the intake of larger feed particles and EE digestibility. The reasons for cows sorting larger feed particles are not clear but might be related to the lower ruminal pH of cows fed BC in comparison with CON. To the best of our knowledge, this is the first study that evaluated ruminal fermentation when feeding an herbal choline. Feeding rumen-protected choline has not altered ruminal pH of cows (Erdman et al., 1984; Mohsen et al., 2011). Although we cannot affirm that BC is inert in the rumen with the data published in literature, authors demonstrated that DM content of BC is half degraded after 18 h of in vitro incubation (Rodríguez-Guerrero et al., 2018). Furthermore, lambs fed increasing doses of BC tended to present a linear increase in plasma phosphatidylcholine concentration (Martínez-Aispuro et al., 2019). Biocholine provides phosphatidylcholine and not free choline, which would make it difficult to estimate the resistance to ruminal degradation since there is synthesis of phosphatidylcholine by ruminal microorganisms and there is no method to separate endogenous from herbal at the duodenal level.

Despite the effects of BC on feed sorting, no treatment differences were observed on DM and nutrient intake. Authors demonstrated that

altering the particle size distribution of TMR may affect feed sorting without altering the actual intake of nutrients (Maulfair et al., 2010). Limited data of DMI and nutrient intake of ruminants fed BC are reported in literature. Similar to observed in the current study, feeding BC has not altered DMI in lambs (Martínez-Aispuro et al., 2019). On the other hand, a meta-analytic study reported that feeding rumen-protected choline to transition cows increases the DMI in both pre- and post-partum periods (Arshad et al., 2020). Few studies evaluated total apparent digestibility of nutrients in lactating cows fed rumen-protected choline. Contrasting with results observed in the current experiment, where only EE digestibility was increased with BC supplementation, Mohsen et al. (2011) reported greater digestibility coefficients of DM, CP, and EE when feeding rumen-protected choline chloride to early lactation cows either at 15 or 30 g/d. In the present study, increased EE digestibility in cows fed BC might be related with its role in digestion, absorption, and transport of fat. Choline-containing compounds stimulate the lipid processing within enterocytes (Koo and Noh, 2001) and phospholipids containing choline are required for the synthesis and release of chylomicrons and VLDL by intestinal villus cells

Table 4

Milk yield and composition, and blood glucose concentration of mid-lactation dairy cows fed increasing levels of biocholine (n = 24).

Item	Treatment ¹ CON	SEM BC7	P-value ² BC14	BC21		L	Q
Yield, kg/d							
Milk*	29.5	30.5	30.0	29.8	0.44	0.722	0.042
ECM ³	33.1	33.8	32.8	33.0	0.35	0.750	0.421
FCM ⁴	32.6	33.3	32.8	33.0	0.33	0.651	0.498
Fat	1.22	1.23	1.21	1.23	0.129	0.756	0.908
Protein	1.00	1.03	1.02	1.01	0.013	0.845	0.089
Lactose	1.50	1.54	1.52	1.52	0.019	0.613	0.091
Composition, %							
Fat	4.19	4.14	4.12	4.21	0.055	0.815	0.095
Protein	3.41	3.40	3.41	3.40	0.112	0.971	0.931
Lactose	5.11	5.09	5.10	5.11	0.016	0.919	0.298
Milk urea nitrogen, mg/dL	13.4	13.7	13.1	13.7	0.212	0.834	0.650
Milk yield ÷ DMI [†]	1.20	1.25	1.21	1.21	0.020	0.606	0.041
FCM ÷ DMI	1.32	1.36	1.33	1.34	0.018	0.885	0.370
ECM ÷ DMI	1.35	1.38	1.35	1.35	0.018	0.747	0.370
Serum glucose, mg/dL	53.9	60.0	57.9	59.9	0.94	0.066	0.295

1Control (CON) and biocholine (Biocholine FC®, Nutriquest Technofeed Nutrição Animal Ltda., Campinas, Brazil) targeting a feeding rate of 7, 14, and 21 g/d (BC7, BC14, and BC21, respectively).

2P-values for contrasts: linear (L), quadratic (Q), and control vs biocholine treatments (CON vs BC).

3Energy-corrected milk (ECM) = (0.327 × milk yield kg/d) + (12.95 × fat yield kg/d) + (7.65 × protein yield kg/d).

4Fat-corrected milk (FCM) = (0.432 + 0.165 × milk fat %) × milk yield kg/d.

*Regression estimates (SE) and quadratic maxima value: intercept = 29.6 (1.535), linear coefficient = 0.124 (0.0570), quadratic coefficient = -0.0057 (0.0026), and maximum value = 10.98 g/d. Coefficients $P \leq 0.033$.

†Regression estimates (SE) and quadratic maxima value: intercept = 1.20 (0.0552), linear coefficient = 0.00556 (0.00301), quadratic coefficient = -0.0003 (0.00014), and maximum value = 9.936 g/d. Coefficients $P \leq 0.0693$.

Table 5

Nitrogen and purine derivatives excretion of mid-lactation dairy cows fed increasing levels of biocholine (n = 24).

Item	Treatment ¹ CON	SEM BC7	P-value ² BC14	BC21		L	Q
Nitrogen, g/d							
Intake	675	672	680	675	9.3	0.838	0.882
Urine	174	179	167	171	7.3	0.726	0.968
Feces	183	190	188	184	4.0	0.997	0.288
Milk	157	162	160	159	2.0	0.719	0.078
Purine derivatives, mmol/d							
Allantoin in urine	158	144	146	145	5.2	0.403	0.474
Uric acid in urine	89.4	78.0	86.4	81.9	3.39	0.626	0.595
Allantoin in milk	77.5	63.1	78.1	79.0	3.29	0.448	0.202

1Control (CON) and biocholine (Biocholine FC®, Nutriquest Technofeed Nutrição Animal Ltda., Campinas, Brazil) targeting a feeding rate of 7, 14, and 21 g/d (BC7, BC14, and BC21, respectively).

2P-values for contrasts: linear (L), quadratic (Q), and control vs biocholine treatments (CON vs BC).

(Kuksis and Mookerjee, 1978). Furthermore, the biliary secretion of lipids (bile acids, phospholipids, and cholesterol) is necessary for cholesterol homeostasis and dietary fat absorption in the small intestine, and biliary phospholipids consist of up to 95% phosphatidylcholine (Alvaro et al., 1986). Authors supplemented rats with choline and observed an increase in bile flow and bile acid secretion rate compared to control (LeBlanc et al., 1998). On the other hand, lactating rat dams fed a choline deficient diet exhibited alterations in intestinal morphology, lower absorptive capacity and impaired lipoprotein secretion from the intestine (da Silva et al., 2015). The improvements in performance of cows fed BC were at least partially related with increased EE digestibility.

The choline source used in this experiment had marginal effects on ruminal fermentation variables but did affect milk yield, suggesting that post-ruminal effects of BC are likely responsible for enhanced performance of cows. No effects were detected on ruminal VFA proportions and concentration suggesting that BC was not able to shift fermentation patterns. The tendency for a quadratic negative effect on ruminal pH caused by BC supplementation has no plausible explanation as described by Sharma and Erdman (1989). Rumen pH has not been altered in cows supplemented with rumen-protected choline (Erdman et al., 1984;

Mohsen et al., 2011). Because we were not aware of any randomized and controlled studies with dairy cows that evaluated the supplementation of the commercial product tested in this study, we fed increasing doses of BC to lactating cows to determine the best dose for improving performance. According to regression analysis, feeding BC at doses around 10 g/d would improve both milk yield and efficiency of milk synthesis in terms of DMI. Mendoza et al. (2020) supplemented fresh cows (from 7 to 67 DIM) with the same BC of the current study at 15 g/d along with 10 g/d of methionine and reported greater milk yield and FCM compared with the control group. It is important to highlight that in the later experiment it is not possible to point out if the positive effects observed in milk yield were either because of BC or methionine supplement (Mendoza et al., 2020). Ewes fed BC at 5 g/d from d -20 until 45 d relative to parturition tended to exhibit greater milk yield in comparison with control group (Alba et al., 2020). There are few studies that evaluated rumen-protected choline supplementation to cows after the peak of lactation. Dietary rumen-protected choline inclusion (up to 0.24% diet DM) linearly increased milk yield without affecting milk composition of mid-lactation cows (151 ± 12 DIM) fed a diet with 16.5% CP (Erdman and Sharma, 1991). These results suggest that choline may be a limiting nutrient not only in periparturient cows but

also in more advanced stages of lactation. Researchers have suggested supplementing rumen-protected choline has increased milk yield due to the provision of methyl-donors, and thus providing higher levels of available methionine (Shahsavari et al., 2016). An indirect effect of choline on general health status have also been pointed out as a possible reason for increases in milk yield (Davidson et al., 2008; Lima et al., 2012). Recently, authors fed BC to lactating ewes and reported increased milk yield and improved colostrum and milk quality (Roque-Jiménez et al., 2020). According to Arshad et al. (2020), the benefits of supplemental choline in postpartum diets are related to an increase on secretion of very low-density lipoprotein and a decrease on the accumulation of reactive oxygen species (Chandler and White, 2017). This meta-analysis used choline as a continuous variable to attempt to model the optimum amount ingested when sufficient treatment means were available, and the amounts of choline supplemented varied substantially (Arshad et al., 2020). Most of the results obtained showed a quadratic effect of choline supplementation, regarding the concentration of fat and lactose in milk, yield of lactose and blood beta hydroxybutyrate concentration, which can suggest that choline presents a limitation to improving the performance of animals (Arshad et al., 2020).

The variables assessed in this study were not able to determine the reasons why BC increased milk yield of cows, and variables related to lipid and choline metabolism, oxidative status, and inflammation should be evaluated in further studies. Bioavailability markers of choline, such as betaine in plasma and milk should be assessed to indicate whether BC increases phosphatidylcholine in the small intestine. Authors demonstrated that betaine in plasma and milk were strongly associated with net free choline portal flux in dairy cows (de Veth et al., 2016). Finally, comparison between studies is complex since the bioavailability of dietary choline in dairy cows is likely to be influenced by factors such as dose, mode of administration (top-dressed vs. incorporated in compound feed), lactation stage (Baldi and Pinotti, 2006), dietary protein (Hartwell et al., 2000), and Met supply (Brusemeister and Sudekum, 2006; Davidson et al., 2008).

Conclusions

Increasing dietary doses of BC had no effect on DM and nutrient intake but tended to quadratically affect OM digestibility and tended to linearly increase EE digestibility. Incrementing doses of BC had minor effects on ruminal fermentation, including a tendency for a negative quadratic effect on rumen pH. Feeding BC to dairy cows may improve milk yield and feed efficiency likely by altering lipid and amino acid metabolism. Regression analysis revealed that the commercial product used in this experiment should be supplemented at doses around 10 g/d to optimize performance of cows. Further studies evaluating the effects of BC on lipid and amino acid metabolism and blood choline metabolites (betaine, homocysteine, etc.) should be performed in mid-lactation cows.

Declaration of Competing Interest

The authors declare no conflicts of interest.

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